

Design FMEA (qualitative, simulation-signal based) — VBF-T1R1FLASH-DFMEA-01

Case: next-generation BEV sedan battery · Generation date: 2026-08-25
Qualitative caliber: S/O three-level ratings based on simulated margin to threshold; RPN = S×O qualitative matrix. Full process/supplier FMEA: N/A (beyond pure simulation boundary).

#	Failure mode	Failure cause	Simulation signal (detectability)	Severity	Occurrence	Design-side mitigation
1	Lithium plating at 4C fast charge	End-of-charge anode surface saturation (electrolyte transport + kinetics)	anode_potential_v min = +0.0114 V (11.4 mV margin)	High	Low (margin > 10 mV; was -77 mV before transport fix)	High-transport electrolyte (σ 5.0 S/m, D 1.2e-9, t_{thin} 0.65); thin collectors; cooling h=150
2	Thermal runaway from fast-charge temperature	Heat generation exceeds cooling at 4C	T_max 323.4 K vs limit 333.15 (9.8 K margin); 367.4 K at h=10	High	Low (margin restored by cooling)	Liquid cooling h ≥ 120 (design point 150); DCR reduced to 20.3 mΩ by high- σ electrolyte
3	Overcharge-induced thermal runaway	0.5C overcharge to 4.7 V pushes cell into SEI-decomposition regime	run-tr triggered=false; overcharge T_max 299.1 K, dT/dt max -7.5e-6 K/s (cooling)	High	Low	Cell mass 38 g (mcp 34.3 J/K), strong cooling h=150, low DCR; pass verified mechanically
4	Electrolyte oxidative decomposition	Voltage window exceedance	Not evaluated (real_compute=false; no HOMO/IE-EA endorsement)	Medium	Not assessed	Honest N/A – requires true-compute endorsement; 4.2 V cutoff within NMC811 stability envelope (domain knowledge)
5	Insufficient capacity / energy	Electrode balance or porosity miscalibration	1C capacity 4.936 Ah (98.7% of nominal); ED 459.4 vs target 392.61	Medium	Low (17% ED margin)	Thick-electrode + thin-collector architecture; validated at 1C DFN

Conclusion

Highest-risk items: plating (mitigated: -77 → +11.4 mV margin via electrolyte transport) and fast-charge temperature (mitigated: cooling + DCR reduction). All mitigations implemented and verified in the final design. Complete FMEA including process/supplier failure modes: N/A (beyond pure simulation boundary).